Vigneshwar Karuppiah Ramanathan



AI & Computer Vision Engineer | ADAS & AD | Data Engineering

+49 15237712988 @ vigneshwar.kr@gmail.com linkedin.com/in/vigneshwar-k-r-b00156177
Full Work Authorization Braunschweig, Germany

SKILLS

Programming & Development: Python, HTML, CSS, JavaScript, SQL, Linux

Computer Vision & Deep Learning: PyTorch, TensorFlow, Keras, OpenCV, Ultralytics (YOLO, SAM), MediaPipe, MVTec HALCON, Object Detection, Image Classification, Semantic & Instance Segmentation, Synthetic Data Generation, Pose Estimation, Camera Calibration

Foundation Models & Generative AI: Hugging Face, Transformers, Vision-Language Models (VLMs), Multimodal Large Language Models (MLLMs), Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search, Prompt Engineering, Agentic systems, LLM-as-a-Judge, Evaluation Metrics

Data Engineering, DevOps & Cloud: Databricks, Microsoft Azure & Azure AI Foundry, MLFlow, MLOps, WandB, MongoDB, Vector DB, REST APIs, GIT, Bitbucket, GitHub, GitLab, JIRA, Confluence

Simulation, CAD, CAE: CARLA, esmini, OpenSCENARIO, OpenDRIVE, OSM, Siemens NX CAD, Catia v5, AutoCAD, Ansys, Scilab

Hardware, Robotics & Computing: CUDA, Slurm, Arduino, RasPi, Industrial Camera Systems, Multi-modal Sensors, Custom KPIs, LaTeX

EXPERIENCE

CARIAD SE (Volkswagen Group's software company), Germany

Master Thesis **Jan 2026 - Aug 2026**

From Fleet Data to Simulation: Automated Scenario Generation using Foundation Models (Written: 1.7, Presentation: 1.3)

- Developed an E2E pipeline to transform sparse fleet data, edge case scenarios into **digital twins** for **virtual validation** of Advanced Driver Assistance Systems (ADAS), Automated Driving (AD) functions
- Leveraged Multimodal Language Models (MLLMs), Vision Language Models (VLMs) for understanding, generation and other downstream tasks
- Implemented **embedding**-based scenario extraction and retrieval to identify edge cases and to enable text-based scenario querying
- Built a **semi-automated** dataset curation pipeline and benchmarked using **multi-metrics evaluation** (lexical similarity, embedding-based metrics and LLM-as-a-Judge approaches)
- **Supervised** an intern for Ground Truth verification workflows
- Investigated **RAG**, **ReAct**-based agentic workflows, visual & textual prompt engineering for automated generation of ASAM OpenDRIVE (.xodr) and OpenSCENARIO (.xosc) simulation files
- Designed an interactive web-based **scenario catalog** for searching, visualizing and editing synthetic data
- Engineered **resource efficient** workflows under computational constraints
- Enabled scenario generation in under **4 minutes per scenario**, including complex topologies and complex interactions

Internship : Jun 2025 - Dec 2025

- Supported the safety department and analyzed customer, production vehicle fleet data
- Developed a **deep learning-based toolkit, dashboard** in **Databricks** to classify true positive vs false positive ADAS/AD activations
- Conducted benchmarking of object detection models and **fine-tuned YOLOv9** on anonymized, preprocessed and imbalanced dataset
- Built a **semi-automated pipeline** for dataset curation, annotation, augmentation, radar perception and incorporated custom KPIs
- **Reduced manual analysis workload by over 98%** through automation of ADAS, AD event analysis
- Successfully **deployed** the solution into the production across **two Volkswagen Group brands** and documented technical details

Infosys Limited

System Engineer: Apr 2021 - Jun 2022

- Worked as a software developer in a Data and Analytics Team - Finance Domain
- Engineered **RESTful backend APIs** using Node.js, Express.js, MongoDB and ensured reliability with **Unit testing** using Mocha, Chai
- Collaborated in building and maintaining **ETL data pipelines** using MS SSMS for providing big data analysis and dashboard generation
- Operated in an agile and scrum environment, maintained technical documentations, organized knowledge sharing and team events

EDUCATION

Technical University of Braunschweig — Nov 2022 - Aug 2026 — GPA
M.Sc. Computational Sciences in Engineering — **1.7** / 4.0

SRM IST, India — Jul 2016 - May 2020 — CGPA
Bachelor of Technology in Mechanical Engineering — **1.8** / 4.0

ADDITIONAL EXPERIENCE

Student Research Assistant, TU Braunschweig, Germany

Institute for Machine Tools and Manufacturing Technology (VaTreBat project) **Feb 2024 - May 2025**

- Contributed to the **autonomous robotics dismantling** process of battery systems
- Researched, integrated suitable hardware and software, and developed prototypes with proof of concept (POC) in **MVTec HALCON**
- Developed image processing, vision, and ML algorithms and trained deep learning models for various battery-related tasks
- Established a multi-sensor environment and interfaced Keyence, KUKA, industrial cameras, calibration systems and microcontrollers
- Collaborated with researchers and industrial partners on prototype development and system integration

Institute for Structural Design **Jun 2024 - Jan 2025**

- Enhanced the existing setup by integrating **Arduino** controlled end effectors within Universal Robots for additive manufacturing

Institute of Flight Guidance (C2Land project) **Jul 2023 - Apr 2024**

- Developed a lightweight rule-based vertipad detection algorithm using **C++** and **OpenCV**, reducing detection latency compared with the existing AI-based approaches
- Structured and populated a vertipad database and implemented visualization based on defined evaluation metrics

Press and Communication, Institute for Mobile Machines and Commercial Vehicles

Simulation Lab, India (Research Internship)

PROJECTS

Specialization Project : Driver Distraction Detection using Hybrid Deep Learning Models (Grade: 1.3)

Niedersächsisches Forschungszentrum Fahrzeugtechnik (NFF), Braunschweig

- Developed a computer vision algorithms, image and video-based hybrid deep learning models for dual camera setup for monitoring **facial distraction** and **hands-on-wheel detection**
- Conducted Exploratory Data Analysis **(EDA)** and evaluated different approaches on benchmark datasets

Institute of Lightweight Systems, DLR, Germany (Grade: 1.0)

Semantic Segmentation of CFRP materials

- Developed a fine-tuned semantic segmentation model to analyze CFRP material images
- Received the **Best Model** and **Best Poster** awards for the final project

TU Braunschweig, Germany

- **AI solutions for remote sensing applications :** Computer Vision tasks, Generative AI, Explainable AI (GitHub)
- **Evaluation of YOLO models on DIOR :** Fine-tuned YOLOv8 outperformed existing models present in DIOR paper by 23% in MAP@0.50 (GitHub)
- **Relevant Courses:** Robot internship, Computer lab Pattern Recognition, Computer Vision and Machine Learning

Other Projects

- **Football AI object detection :** Python, OpenCV, YOLO, scikit-learn (GitHub)
- **Instance Segmentation for Robotic bin picking :** Python, OpenCV, Ultralytics: YOLO + SAM 2
- **Object detection of Battery system components :** Python, OpenCV, rule-based algorithms, YOLO
- **Simulation of optimal 3D environment mapping with dual axially rotating Lidar :** Scilab, Point cloud analysis

LANGUAGES, TEST SCORES

English	Full Professional Proficiency	**German (B2)**	Intermediate Proficiency	**Deutsch DSH**	DSH 1 (B2)

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Microsoft Certified: Azure Fundamentals

Machine Learning Specialization

Improving Deep Neural Networks

Introduction to MCP, Agent Skills from Anthropic

FIND ME

Portfolio Website
https://vigneshwar.my.canva.site/

GitHub
github.com/Vigneshwar-KR